SPOILER ALERT

This book is an excuse for me to dig deep into the question of where it's all going, what that all means, and what we can learn about the universe we live in by asking these questions. There isn't just one accepted answer to any of this—the question of the fate of all existence is still an open one, and an area of active research in which the conclusions we draw can change drastically in response to very small tweaks in our interpretations of the data. In this book, we'll explore five possibilities, chosen based on their prominence in ongoing discussions among professional cosmologists, and dig into the best current evidence for or against each of them.

Each scenario presents a very different style of apocalypse, with a different physical process governing it, but they all agree on one thing: there will be an end. In all my readings, I have not yet found a serious suggestion in the current cosmological literature that the universe could persist, unchanged, forever. At the very least, there will be a transition that for all intents and purposes destroys *everything*, rendering at least the observable parts of the cosmos uninhabitable to any organized structure. For this purpose, I will call that an ending (with apologies to any temporarily sentient bursts of random quantum fluctuation[fn8] that may be reading this). A few of the scenarios carry with them a hint of possibility that the cosmos might renew itself, or even repeat, in one way or another, but whether some tenuous memory of previous iterations can persist in any way is a matter of rather intense ongoing debate, as is whether or not anything like an escape from a cosmic apocalypse could in principle be possible. What seems most likely is that the end for our little island of existence known as the observable universe is, truly, the end. I'm here to tell you, among other things, how that might happen.

Just to get everyone on the same page, we'll start with a quick catch-up on the universe from the beginning until now. Then we'll get on with the destruction. In each of five chapters, we'll explore a different possibility for the end, how it might come about, what it would look like, and how our changing knowledge of the physics of reality leads us from one hypothesis to another. We'll start with the Big Crunch, the spectacular collapse of the universe that would occur if our current cosmic expansion were to reverse course. Then come two chapters of dark-energy-driven apocalypses, one in which the universe expands forever, slowly emptying and darkening, and